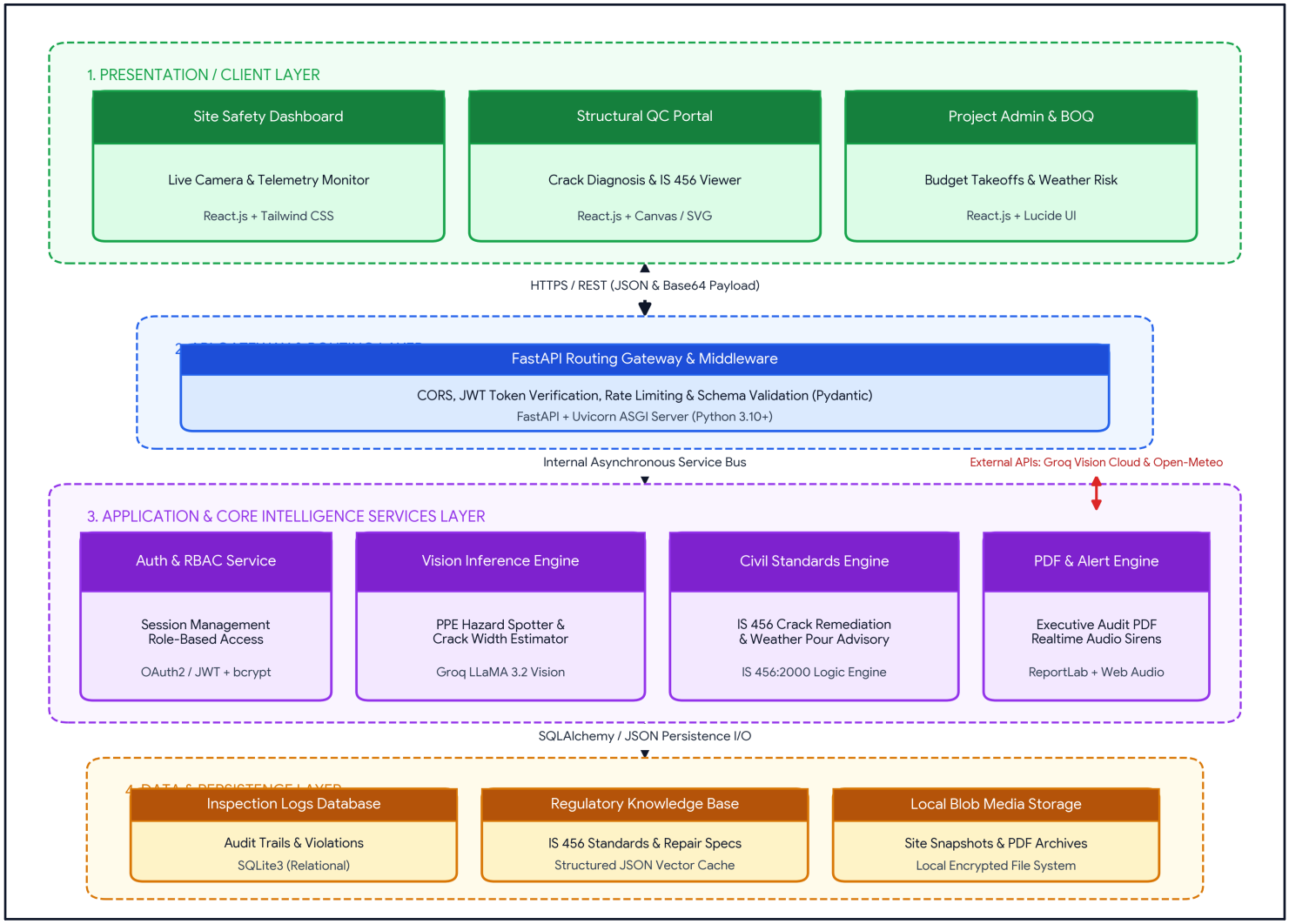
Solution Architecture

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence (CONSTRUCTAI)	**Maximum Marks:**	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. The multi-tiered architecture diagram below maps out the **Presentation Layer**, **API Gateway / Load Balancer**, **Application / Microservices Layer**, and **Data Layer** tailored for CONSTRUCTAI.



Solution Architecture

Component Description Table

Comprehensive technical breakdown of each architectural layer, detailing role specifications, asynchronous throughput, and selected open-source & cloud technologies.

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer [Frontend Tier]	Delivers an intuitive, single-page application (SPA) with responsive high-contrast views designed for outdoor sunlight visibility on tablets and smartphones. • **Features:** Drag-and-drop CCTV/drone snapshot upload, interactive bounding box overlays for missing PPE, visual crack severity gauges, and instant one-click PDF export.	• **React.js 18** (Vite) • **Tailwind CSS** • **Lucide React Icons** • **HTML5 Canvas / SVG** • **Axios HTTP Client**
API Gateway & Middleware Layer [Routing Tier]	Acts as the unified entry point for all client requests. Manages cross-origin resource sharing (CORS), client request rate limiting, payload base64 integrity validation, and secure session JWT verification before dispatching to backend microservices.	• **FastAPI** (Python 3.10+) • **Uvicorn ASGI Server** • **Pydantic v2 Models** • **python-jose (JWT)**
Application & Microservices Layer [Core AI Tier]	Executes core business logic and multimodal vision intelligence across dedicated modular services: • **Vision Inference Engine:** Dispatches site imagery to Groq Cloud LPUs running LLaMA 3.2 Vision for sub-second PPE violation and crack width detection. • **Civil Standards Engine:** Cross-references visual crack metrics against **IS 456:2000 Section 35.3.2** to output chemical repair specs. • **Weather & Alert Engine:** Correlates Doppler radar with concrete pour schedules and sounds real-time audible warning sirens.	• **Groq LLaMA 3.2 Vision** (11B/90B API) • **Open-Meteo Weather API** • **ReportLab PDF Engine** • **Pillow (PIL)** • **Python Asyncio Worker Pool**
Data & Persistence Layer [Storage Tier]	Provides persistent relational storage and fast in-memory caching: • **Inspection Logs:** Stores timestamped safety violation events, worker counts, and compliance scores. • **Standards Repository:** Pre-indexed JSON knowledge base of IS 456 and National Building Code guidelines. • **Media Storage:** Local encrypted repository archiving site imagery and generated executive audit reports.	• **SQLite3** (Relational DB) • **SQLAlchemy ORM** • **Encrypted File System** • **Structured JSON Vector Store**